RETAIL INVENTORY & DEMAND ANALYSIS

Business Analysis Report | Python

Project purpose
Explore retail sales, revenue, inventory, demand, product and regional performance using Python. Translate the observed patterns into practical business recommendations.

Dataset characteristic	Details
Dataset	Retail Store Inventory Forecasting Dataset
Records and fields	76,000 records; 16 original fields
Analysis period	1 January 2022 – 30 January 2024
Stores / products	5 stores / 20 products
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn
Data nature	Synthetic dataset (as identified for this project)

Prepared from the executed Google Colab notebook supplied for review. Numerical findings in this report reflect the saved notebook outputs.

1. Executive Summary

The notebook analyses 76,000 retail observations across five stores, 20 products, five categories and four regions. The analysis covers data quality, engineered revenue and inventory indicators, exploratory analysis, seasonal patterns, promotional comparisons and statistical relationships.

Headline metric	Result
Total net revenue	412,590,589.82
Total units sold	6,750,876
Total recorded demand	7,928,104
Average listed price	67.73
Average discount	9.09%
Potential-shortage records	10,394 (13.68%)
Inventory-surplus records	65,405 (86.06%)

Overall interpretation: Groceries is the largest revenue category, North is the highest-revenue region, S003 is the highest-revenue store, and Summer has the highest average demand. The inventory comparison flags 13.68% of records where recorded demand exceeds recorded inventory. These are analytical indicators, not confirmed lost sales or stockouts.

2. Business Objectives

The analysis was designed to: (1) understand revenue and sales trends; (2) identify high- and low-performing products and categories; (3) compare regional and store performance; (4) examine inventory relative to recorded demand; (5) explore seasonal, promotional, discount and competitor-price patterns; and (6) produce evidence-based recommendations.

3. Dataset and Methodology

The original dataset contains Date, Store ID, Product ID, Category, Region, Inventory Level, Units Sold, Units Ordered, Price, Discount, Weather Condition, Promotion, Competitor Pricing, Seasonality, Epidemic and Demand. The notebook converted Date to datetime and derived Year, Month and Day. It created Gross Revenue, Discount Amount, Net Revenue, Inventory Gap, Potential Shortfall, Inventory Surplus and Inventory Status, plus Month Name.

Revenue calculation: Gross Revenue = Units Sold × Price. Discount Amount = Gross Revenue × Discount / 100. Net Revenue = Gross Revenue – Discount Amount. This assumes Discount is expressed as a percentage, consistent with the observed values 0, 5, 10, 15, 20 and 25.

Inventory comparison: Inventory Gap = Inventory Level – Demand. Potential Shortfall = max(Demand – Inventory Level, 0). Inventory Surplus = max(Inventory Level – Demand, 0). This is a row-level comparison between two dataset fields; it does not establish whether the inventory was available at the precise time of demand.

4. Data Quality and Preprocessing

Check	Notebook result	Assessment
Dataset dimensions after cleaning	76,000 × 19	Date features added; no rows removed
Missing values	0	No missing values reported
Exact duplicate rows	0	No exact duplicates found
Invalid date conversions	0	All dates converted successfully
Negative values in key business fields	0	No negatives found in tested fields
Discount values	0, 5, 10, 15, 20, 25	Consistent with percentage points
Potential outliers	Detected in 7 numeric fields	Retained; not assumed to be errors

The IQR method flagged 2,759 Inventory Level records, 1,411 Units Sold records, 7,524 Units Ordered records, 70 Price records, 12,413 Discount records, 185 Competitor Pricing records and 986 Demand records. These were retained because unusual values may be valid retail observations. In particular, discounts of 20% and 25% are valid observed discount levels even though the IQR rule flags them.

Data-quality caution: Units Ordered has a median of zero and a 75th percentile of zero, with a maximum of 1,616. This highly concentrated distribution should be investigated before making operational conclusions about replenishment.

5. Overall Business Performance

Total net revenue across the 76,000 observations is 412.59 million currency units. Total recorded units sold are 6.75 million, compared with total recorded demand of 7.93 million. The dataset does not specify a currency; therefore, monetary figures are presented in unspecified currency units.

6. Revenue Trends

Period	Net revenue	Interpretation
2022	201.23M	Full calendar year in dataset
2023	196.15M	2.53% lower than 2022
2024	15.21M	Only 1–30 January; partial year

The notebook identifies August 2022 as the highest-revenue month (approximately 19.98M) and February 2023 as the lowest-revenue month (approximately 9.75M). Monthly values should be compared with care because the 2024 period is incomplete. The notebook's calculated –92.25% change for 2024 is not a meaningful full-year decline; it compares one month with a full prior year. The valid full-year comparison available is the 2.53% decrease from 2022 to 2023.

7. Category Performance

Category	Net revenue	Revenue share	Units sold
Groceries	152.96M	37.07%	3,127,335
Furniture	100.89M	24.45%	880,654
Clothing	77.25M	18.72%	1,150,873
Electronics	59.83M	14.50%	757,335
Toys	21.66M	5.25%	834,679

Insight: Groceries contributes the largest revenue share and also has the highest unit sales and recorded demand. Furniture ranks second in revenue despite fewer units sold, consistent with its much higher average price (121.51). Toys has the lowest revenue contribution. Category performance therefore reflects both sales volume and price levels.

8. Product Performance

Product	Net revenue	Units sold	Observation
P0014	33.54M	317,397	Highest revenue
P0011	30.44M	332,883	Second-highest revenue
P0009	29.52M	376,771	Third-highest revenue
P0007	12.97M	379,709	Highest units sold; lowest revenue
P0001	12.67M	356,814	Lowest revenue

The product with the highest sales volume is not the highest-revenue product. P0007 sells the most units but generates 12.97M in net revenue, while P0014 generates 33.54M from fewer units. Product prioritisation should therefore consider revenue, unit volume, demand and inventory together.

9. Regional Performance

Region	Net revenue	Revenue share
North	154.14M	37.36%
East	89.26M	21.63%
West	85.60M	20.75%
South	83.59M	20.26%

North is the strongest region by revenue, unit sales and demand. South has the lowest revenue contribution. The average discount levels across regions are very similar (approximately 9.08%–9.10%), so discount differences alone do not explain the observed regional revenue gap.

10. Store Performance

Store	Net revenue	Revenue share	Average inventory
S003	89.26M	21.63%	321.99
S004	85.60M	20.75%	293.24
S002	83.59M	20.26%	332.01
S001	81.56M	19.77%	277.66
S005	72.59M	17.59%	280.40

S003 is the highest-revenue store and S005 is the lowest. However, S005's unit sales are close to other stores, suggesting that revenue differences should be explored alongside product mix and price rather than interpreted as a simple difference in sales volume.

11. Inventory and Demand

Inventory status	Records	Share
Inventory surplus	65,405	86.06%
Potential shortage	10,394	13.68%
Inventory balanced	201	0.26%

The category-level cumulative potential shortfall is highest for Groceries (222,394), followed by Clothing (100,926), Toys (64,146), Electronics (53,034) and Furniture (35,448). Groceries should be a priority for further investigation because it combines the highest average demand (120.98) with the largest cumulative potential shortfall.

The five products with the largest cumulative potential shortfall are P0008 (43,554), P0018 (36,213), P0009 (34,354), P0010 (31,060) and P0002 (29,030). These figures are summed row-level gaps, not verified quantities of lost sales.

12. Seasonal and Monthly Demand

Season	Average demand	Total demand
Summer	112.86	2,076,539
Winter	103.41	2,171,530
Autumn	103.42	1,882,298
Spring	97.70	1,797,737

Summer has the highest average demand (112.86), while Spring has the lowest (97.70). August has the highest average monthly demand (119.80), followed by June (117.35). Note that the notebook's monthly demand table groups all years together by month number, so these are pooled month-of-year averages rather than a year-by-year seasonal comparison.

13. Promotions and Discounts

Promotion flag	Average units sold	Average demand	Average discount
No promotion (0)	81.83	95.03	4.98%
Promotion (1)	103.10	123.27	17.46%

Records with a promotion show higher average units sold and average demand. This comparison is observational: promotional records also have much higher average discounts, and other factors may differ. It does not prove that promotions caused the higher sales or demand.

Across discount levels, average units sold is approximately 81.57 at 0% discount, 81.69 at 5%, 87.66 at 10%, 102.88 at 15%, 103.77 at 20% and 103.59 at 25%. Average revenue per record does not rise consistently with discount level. The business should evaluate margin and profitability before recommending deeper discounts; the dataset does not include cost or gross margin.

14. Competitor Pricing

Price position	Records	Average units sold	Average demand
Above competitor	38,140	88.74	104.35
Below competitor	37,823	88.92	104.29
Same price	37	80.95	97.27

Average units sold and demand are almost identical for records priced above and below competitors. The analysis does not show a meaningful descriptive difference between those two groups. Only 37 records have exactly the same price as the competitor, so that group is too small for strong conclusions.

15. Correlation and Statistical Findings

Variable	Correlation with demand
Units Sold	+0.833
Units Ordered	+0.512
Gross Revenue	+0.498
Net Revenue	+0.470
Discount	+0.225

Variable	Correlation with demand
Inventory Level	+0.127
Price	–0.023
Competitor Pricing	–0.023

Units Sold has the strongest positive linear association with Demand. Units Ordered also has a moderate positive association. Price and competitor pricing have correlations close to zero with Demand in this dataset. Correlation is not causation, and some revenue relationships are partly mathematical because revenue is calculated using Units Sold and Price.

The statistical summary shows right-skewed distributions for Inventory Level (skew 1.59), Units Ordered (2.52) and Net Revenue (1.50). Units Ordered also has high kurtosis (7.31), indicating a heavy-tailed distribution relative to a normal distribution. This supports further investigation of its many zero values and extreme high values.

16. Key Business Recommendations

1. Prioritise grocery replenishment analysis. Groceries contributes 37.07% of revenue and has the highest category-level cumulative potential shortfall. Review product-level demand, inventory availability and replenishment timing for this category.

2. Investigate the products with the largest potential shortfall. Start with P0008, P0018, P0009, P0010 and P0002. Check whether these patterns are concentrated in particular stores, regions or dates before changing order quantities.

3. Review inventory allocation. Although 86.06% of records are classified as surplus under the current comparison, 13.68% are flagged as potential shortages. Consider whether stock can be redistributed across stores or products, subject to actual logistics and service constraints.

4. Plan for seasonal demand variation. Summer has the highest average demand, with August and June showing the highest pooled monthly averages. Use these as signals for further planning, but validate with store- and product-level patterns.

5. Evaluate promotions using profitability, not units alone. Promotional records show higher average units sold, but the dataset lacks product cost and margin. Add cost data before concluding whether promotions improve profit.

6. Review the lowest-revenue store and category mix. S005 and Toys have the lowest revenue results. Compare their product mix, prices, demand and inventory before proposing corrective action.

7. Investigate the Units Ordered field. Its median and 75th percentile are zero, while its maximum is 1,616. Confirm the business definition and data-generation logic before using it for replenishment conclusions.

17. Limitations and Further Work

• The dataset is synthetic; findings describe this dataset and should not be presented as verified real-world retailer performance.
• The currency is unspecified.
• Demand is a supplied field; its definition and relationship to Units Sold should be confirmed.
• Inventory Gap compares Inventory Level with Demand but does not model timing, receipts, safety stock, lead times or lost sales.
• The analysis is descriptive and correlational, not causal.
• 2024 includes only January 1–30; annual comparisons should use full years only.
• Weather Condition and Epidemic were not analysed in the notebook outputs reviewed.
• No forecasting or predictive machine-learning model has been built.

Recommended additional analyses before final submission

1. Explore Weather Condition and Epidemic against average demand and units sold.
2. Investigate Units Ordered zeros and extreme values by date, product and store.
3. Compare inventory status by product, store, category and region.
4. Verify the dataset grain and uniqueness of Date + Store ID + Product ID.
5. Add profit or cost data if profitability analysis is required.
6. If forecasting is an objective, build and validate a separate time-series or predictive model; do not describe the current descriptive analysis as forecasting.

18. Conclusion

The Python analysis provides a broad descriptive view of retail revenue, product and category performance, regional and store contributions, inventory-demand gaps, seasonal patterns, promotions and pricing relationships. The strongest findings are the contribution of Groceries and the North region, the difference between the highest-volume and highest-revenue products, and the presence of potential inventory shortfalls in 13.68% of records. These findings can guide further investigation into product replenishment, inventory allocation and promotional strategy. The project should be described as a descriptive retail analytics study using synthetic data, not as a validated demand-forecasting system.

Report prepared from the supplied notebook's executed outputs. Figures may be rounded for presentation; calculations should be re-run from the source dataset if the notebook is modified.

Appendix: Selected Business Charts

These charts are generated from the numerical results saved in the supplied notebook. They complement the written findings in the main report. Figures are descriptive and should not be interpreted as causal evidence.

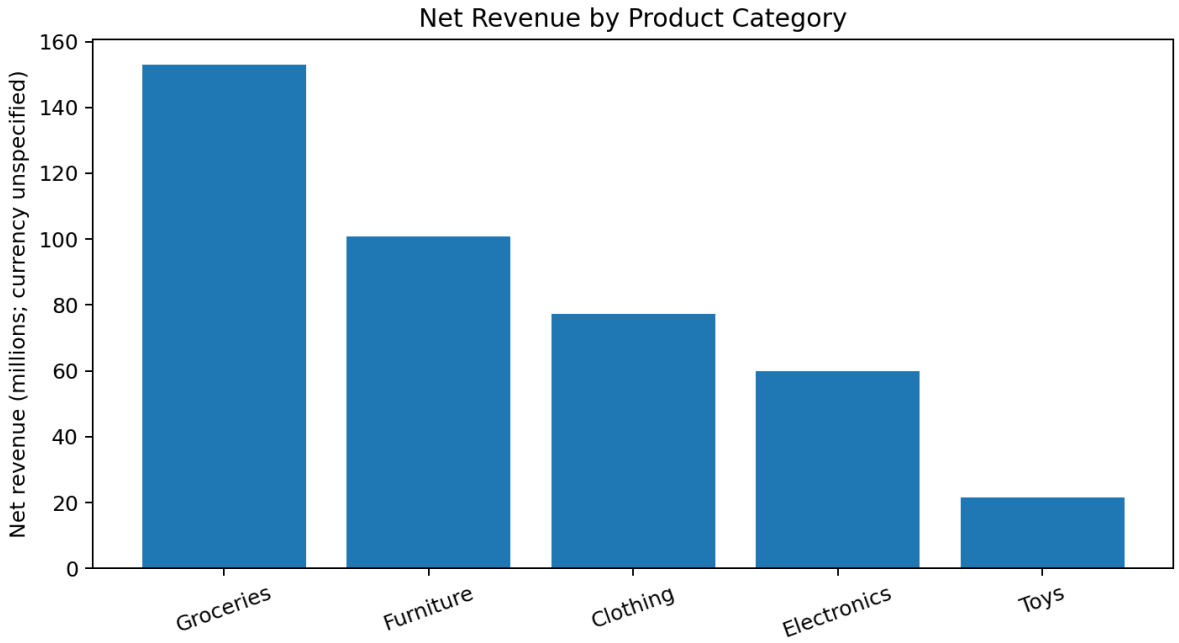


Figure A1. Category revenue Groceries contributes the largest share of net revenue, while Toys contributes the least.

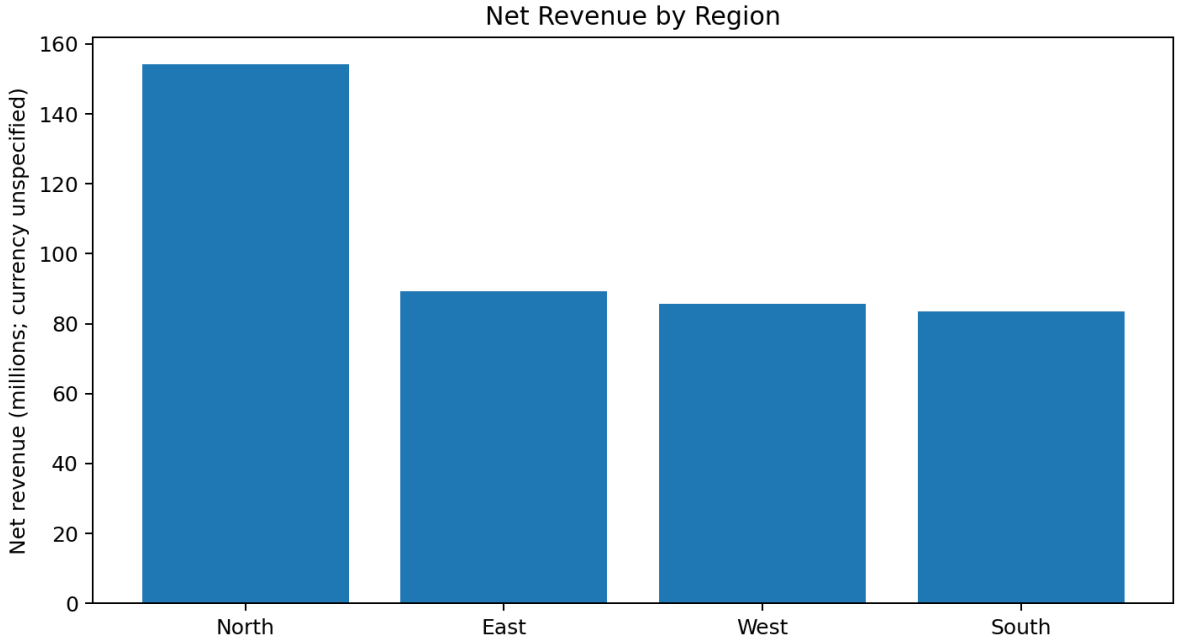


Figure A2. Regional revenue North is the highest-revenue region in the analysed dataset.

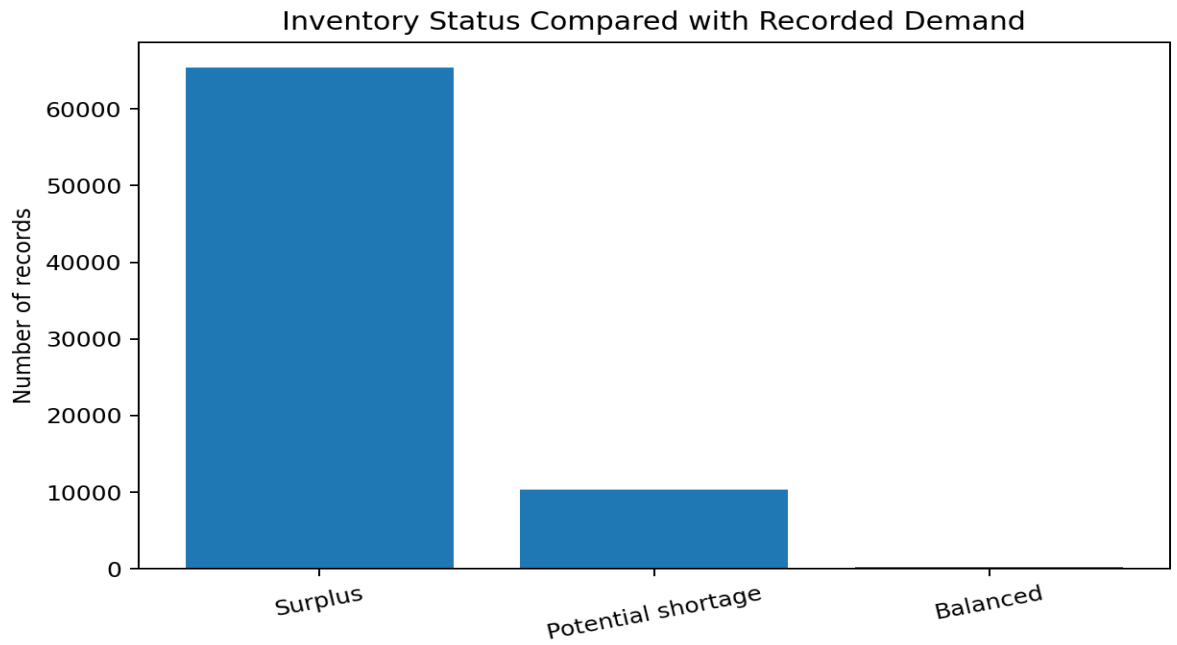


Figure A3. Inventory status 13.68% of records show inventory below recorded demand; this is an indicative gap, not confirmed lost sales.

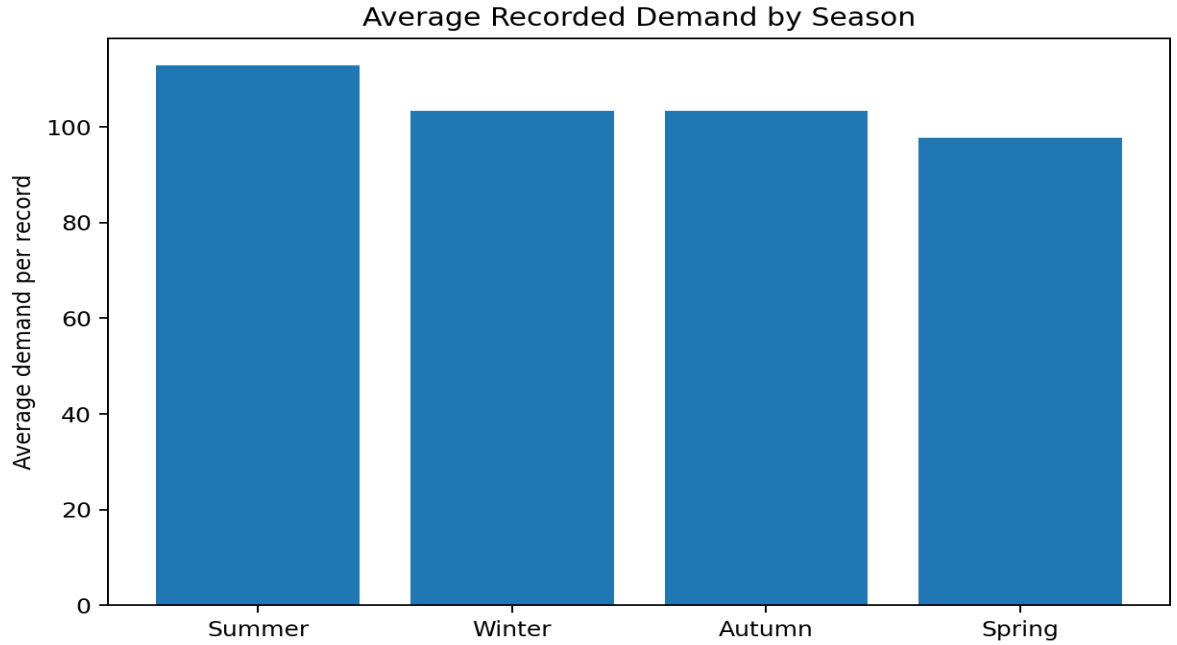


Figure A4. Average demand by season Summer has the highest average recorded demand; Spring has the lowest.

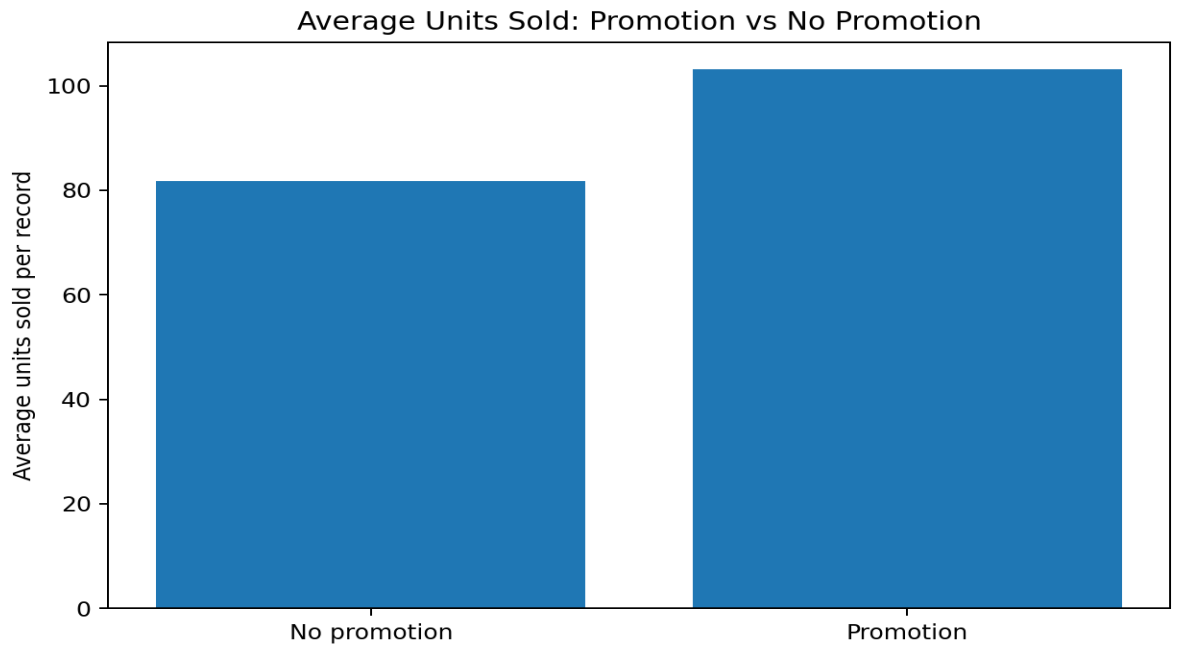


Figure A5. Promotion comparison Promotional records have higher average units sold, but this observational comparison does not prove a causal promotion effect.

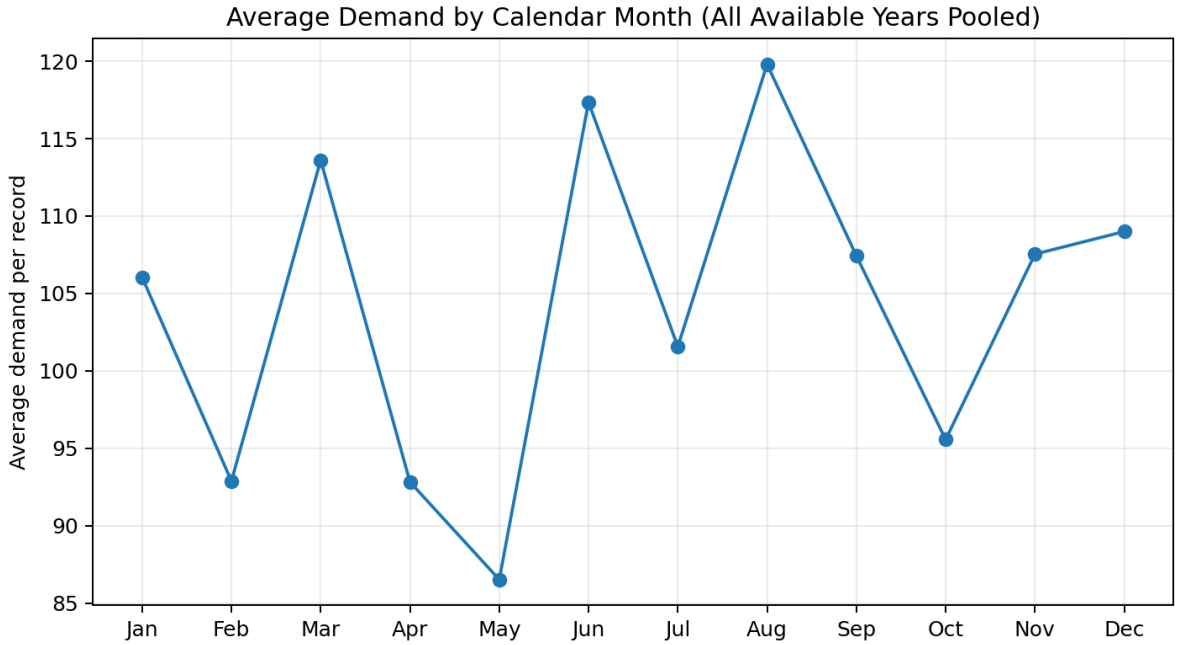


Figure A6. Average demand by month August has the highest pooled average demand. Month values combine available years and should not be read as a year-specific trend.